4.05 Further Algebra			
4.05a	Roots of equations	a) Understand and be able to use the relationships between the symmetric functions of the roots of polynomial equations and the coefficients. *Up to, and including, quartic equations.* *e.g. For the quadratic equation* $ax^2 + bx + c = 0$ *with roots* α *and* β, $\alpha + \beta = -\frac{b}{a}$ *and* $\alpha\beta = \frac{c}{a}$.	
4.05b	Transformation of equations	b) Be able to use a substitution to obtain an equation whose roots are related to those of the original equation. *Equations will be of at least cubic degree.*	
4.05c	Partial fractions		c) Extend their knowledge of partial fractions up to rational functions in which the denominator may include quadratic factors of the form $ax^2 + c$ for $c >$ and in which the degree of the numerator may be equal to, or exceed, the degree of the denominator. *See H240 section 1.02y.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.